Adaptive Multi-Site Stimulation Control Using RL in an MDP

Learning sequential stimulation policies under state-dependent EEG response dynamics

Fatih Karatay and Cody Moxam
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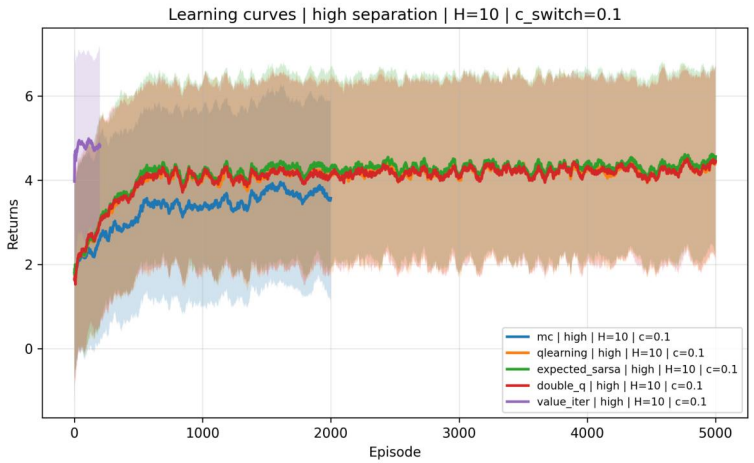
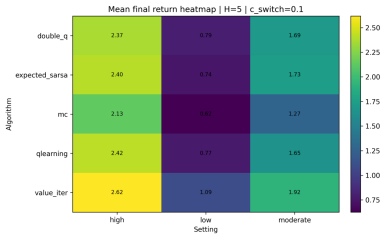
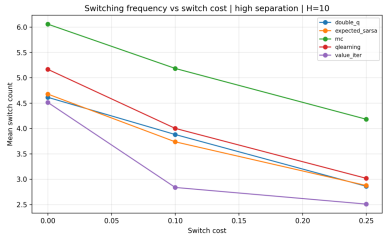






Problem framing

Why this is a reinforcement learning problem rather than a bandit

- The agent selects among four stimulation sites S1-sites S1-S4 over a finite horizon.
- Immediate reward depends on the current patient responsiveness state and the EEG response it induces.
- Responsiveness changes over time: repeated stimulation can push the patient from baseline to receptive to non-receptive.
- Because actions change future state occupancy and future reward, the task is sequential control.

Core idea

Optimal behavior is not simply “choose the best site.”
It is “sequence sites to accumulate reward while avoiding state degradation.”

State flow

baseline → receptive → non-receptive

switching promotes recovery

Representative reward logic

favorable = +1 | **neutral** = 0 | **unfavorable** = -1 |
switch cost ∈ {0, 0.1, 0.25}

MDP formulation and experimental design

Finite-horizon, fully observed environment with 90 total configurations

MDP components

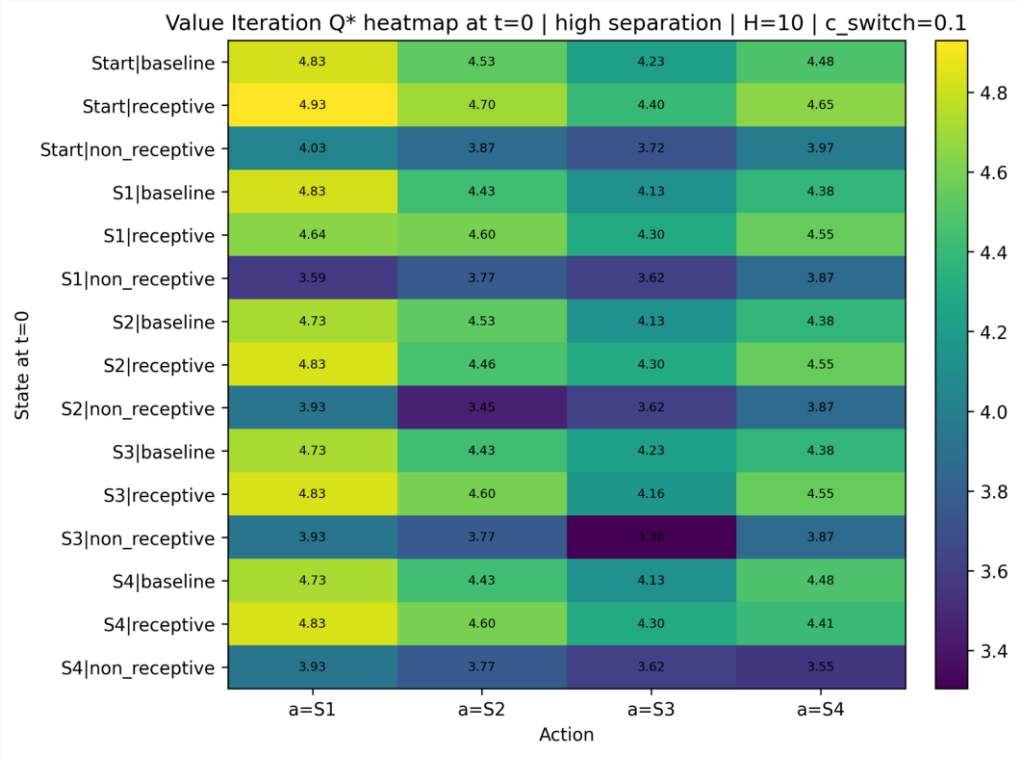
components

State	(current site, patient state, t)
Actions	{S1, S2, S3, S4}
Patient states	{baseline, receptive, non-receptive}
Observation	EEG ∈ {favorable, neutral, unfavorable}
Reward	+1 / 0 / -1 with optional switch cost

Experiment grid

grid

- Separation settings: high, moderate, low
- Episode horizons: H = 5 and H = 10
- Switch cost: 0, 0.1, 0.25
- Algorithms: MC, Q-Learning, Expected SARSA, Double Q, Value Iteration
- Model-free results averaged over 10 random seeds



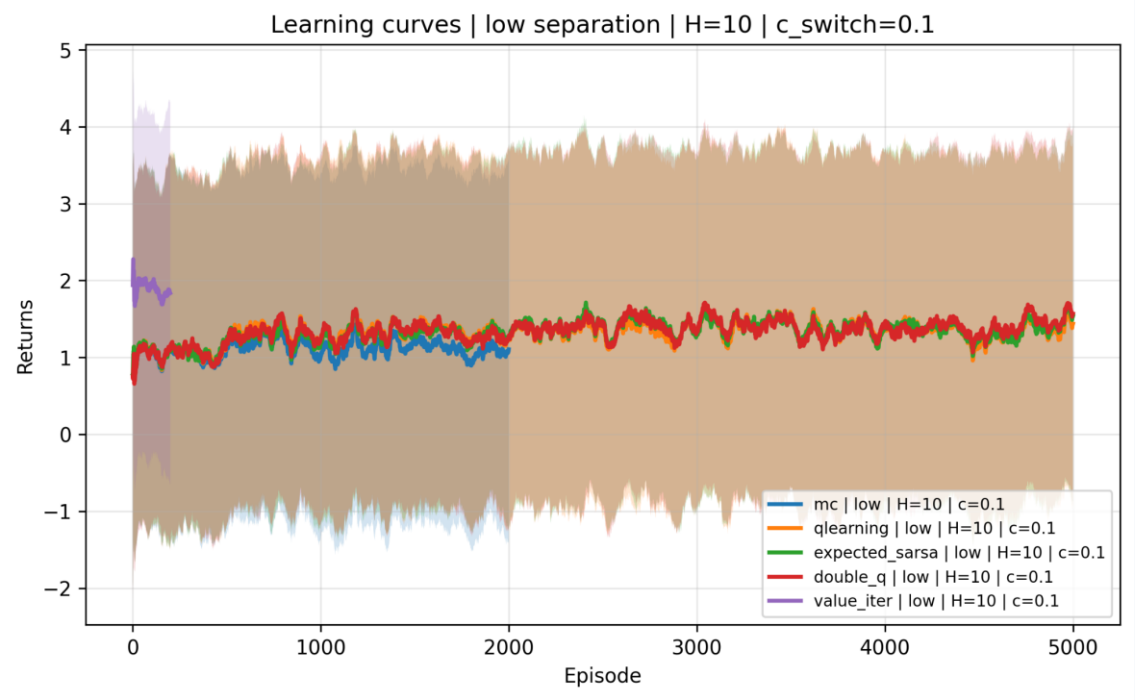
Q at t=0 (high separation, H=10, c_switch=0.1): S1 is usually best, with S4 as a secondary option.*

Learning performance improves with reward separability separability

H1

Representative learning curves for H=10 and c_switch=0.1

High separation: TD methods converge quickly to strong positive return; MC lags; Value Iteration is the Iteration is the upper bound.

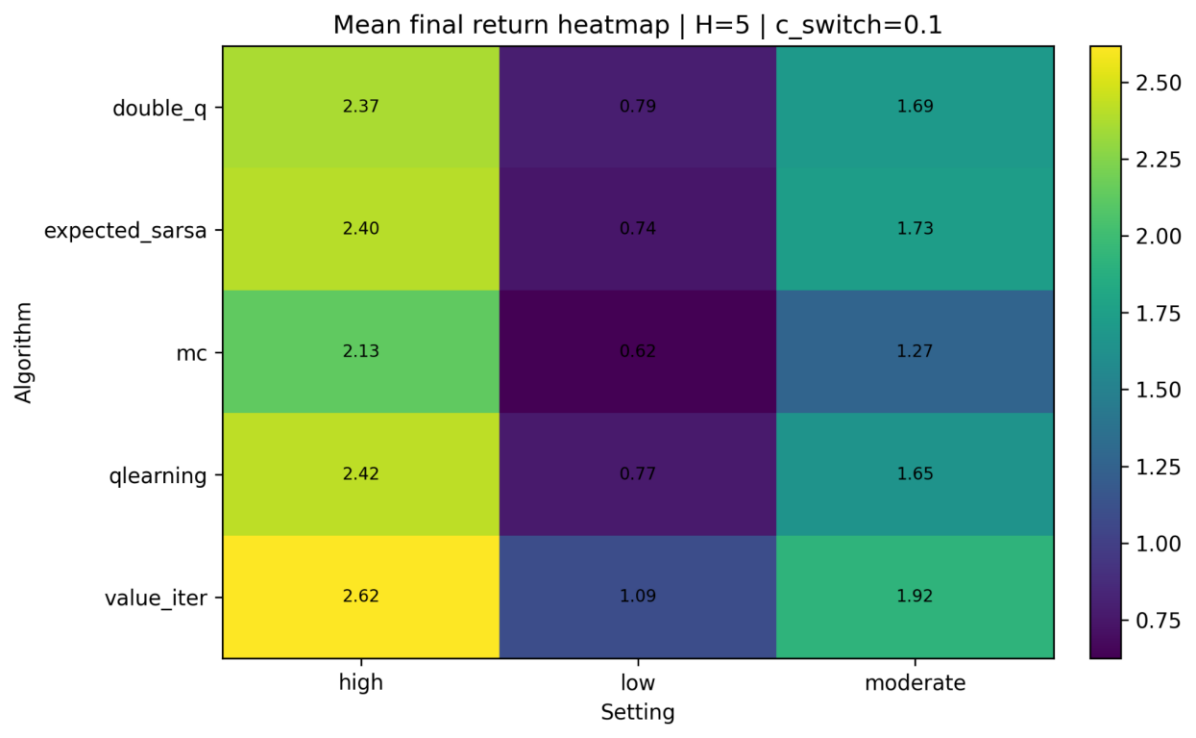


Low separation: all methods converge more slowly and to lower final return.

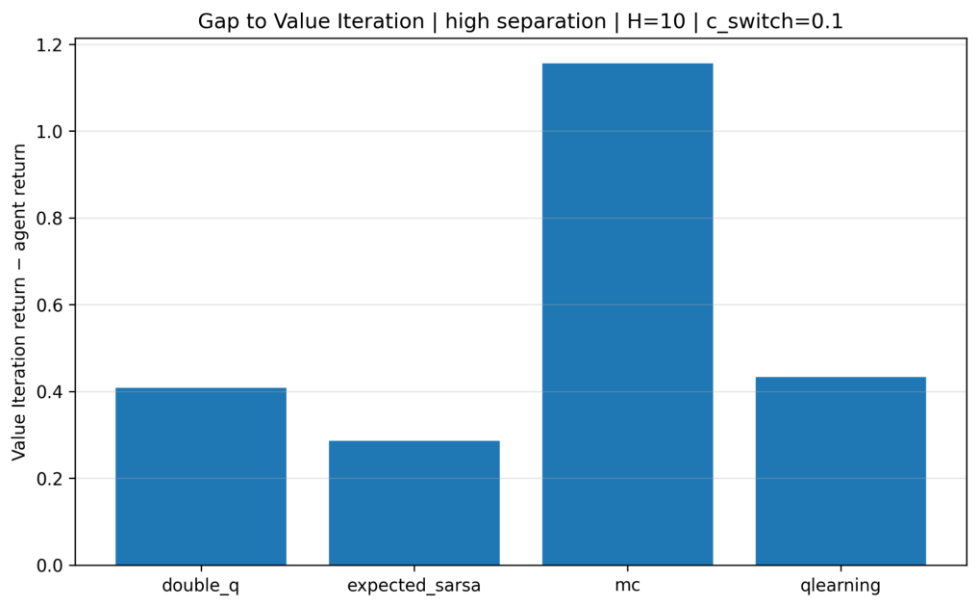
- High separation produces faster convergence and substantially higher return than low separation.
- Expected SARSA, Q-Learning, and Double Q cluster closely in the high-separation regime.
- This supports H1: clearer reward structure improves learnability.

Cross-setting performance and gap to optimal

Heatmap at H=5, c_switch=0.1 and benchmark gap at high separation



Mean final return heatmap. High separation is easiest; low separation is hardest; Value Iteration is best overall.

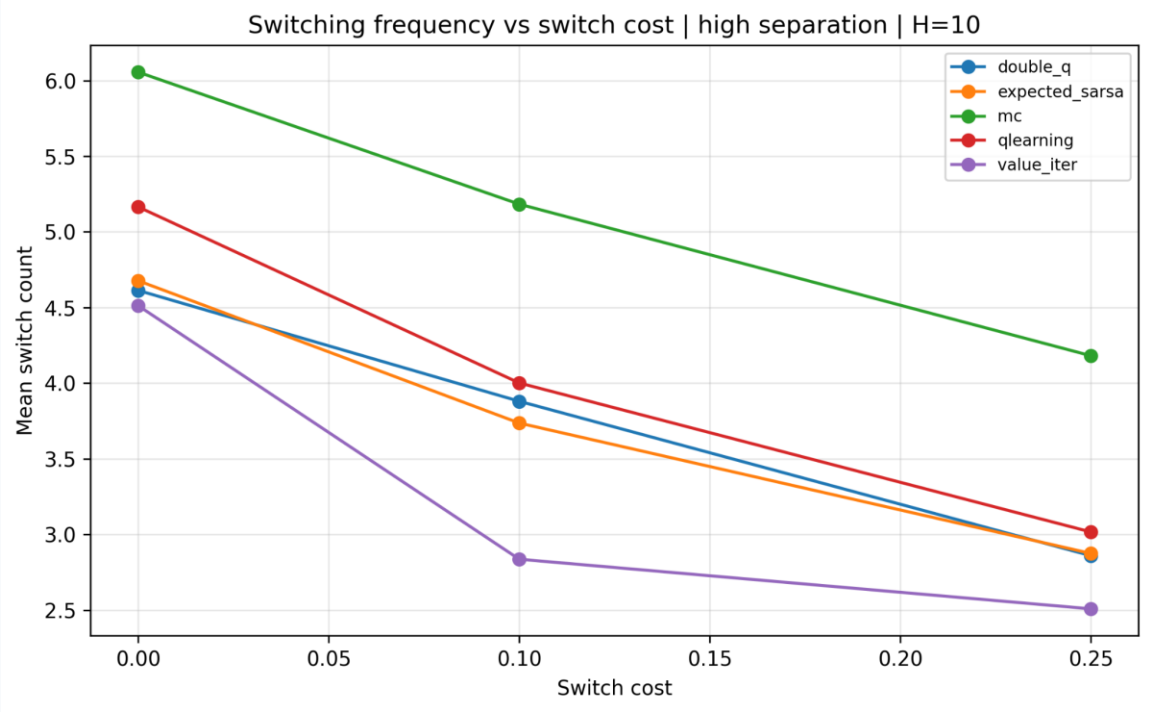


Gap to Value Iteration (high separation, H=10, c_switch=0.1). Expected SARSA is closest to optimal in this representative setting.

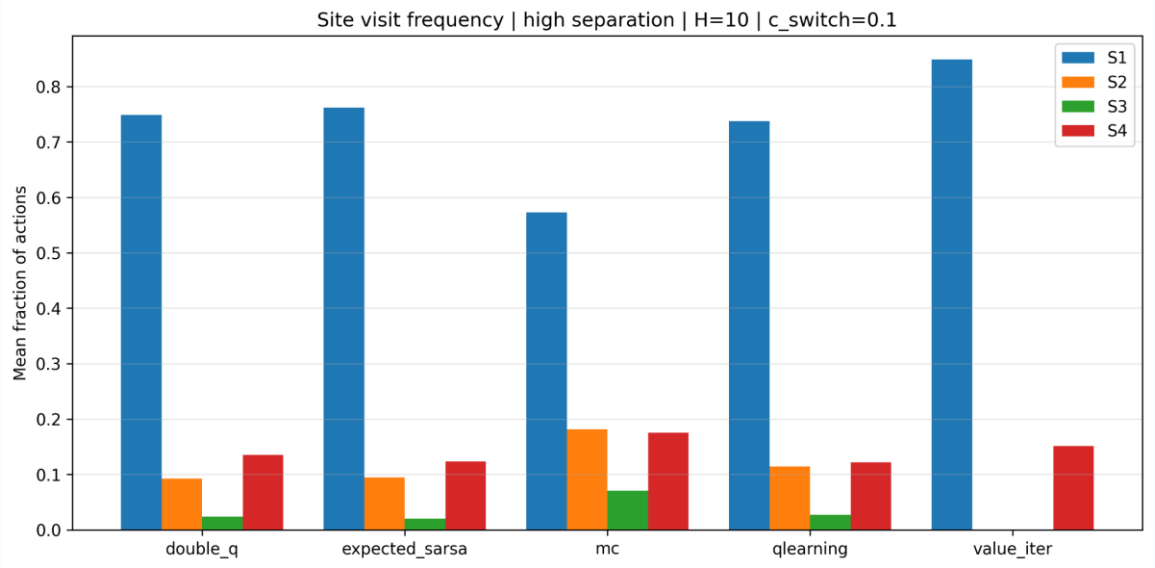
- In the H=5 heatmap, Value Iteration reaches 2.62 in the high setting versus 1.09 in the low setting.
- Among model-free methods, Q-Learning and Expected SARSA are strongest in high separation.
- The non-zero optimality gaps quantify how closely model-free agents approach the optimal policy.

Switching cost shapes policy behavior

Representative high-separation results at H=10



Mean switch count declines as c_switch increases for every algorithm.

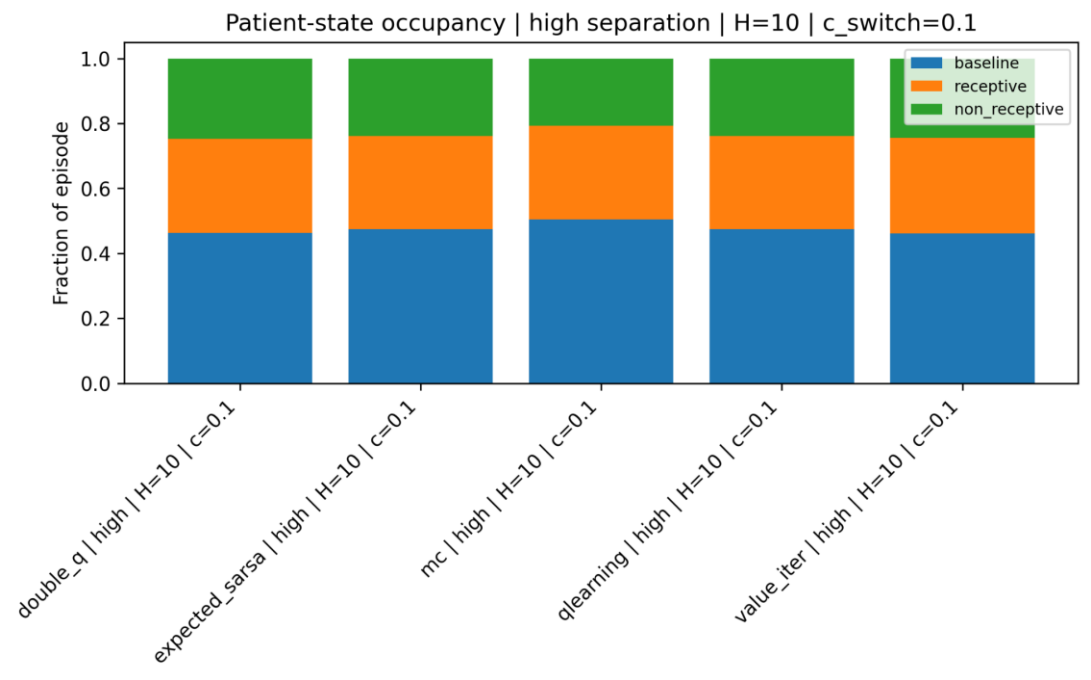


High-separation policies concentrate on S1, with smaller allocations to S2-S4.

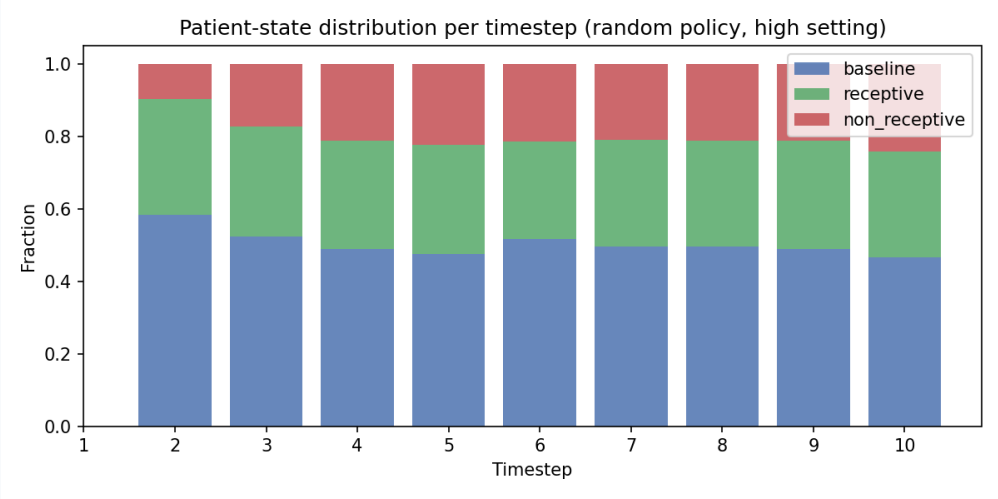
- Higher switching cost discourages unnecessary site changes.
- Policies still switch when needed to avoid pushing the patient into the non-receptive state.
- This supports H2 and shows that the learned behavior is cost-sensitive, not myopic.

Agents learn state-aware sequential behavior

Occupancy patterns help explain how reward is sustained over time



Across algorithms, most time is spent in baseline and receptive states, with a smaller fraction in the non-receptive state.

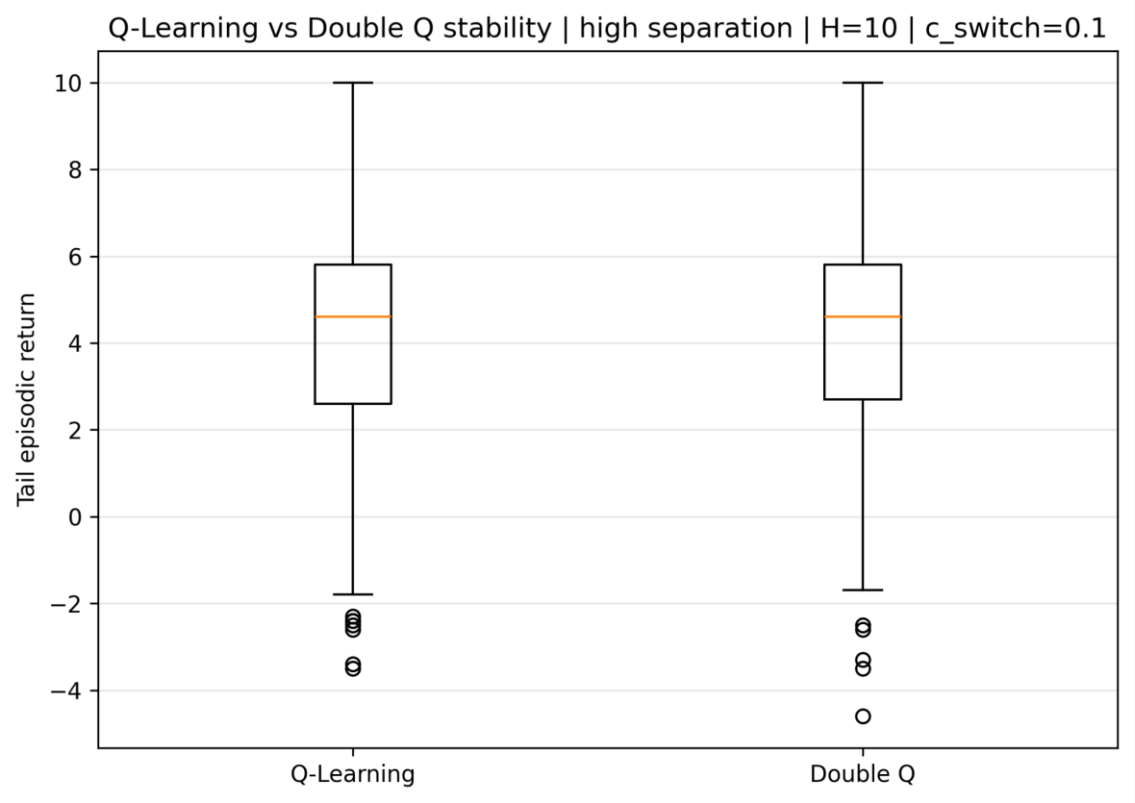


Sanity check under a random policy: all three patient states remain occupied over time, confirming nontrivial dynamics.

- Good policies do not simply hammer one site indefinitely.
- They manage the state trajectory: exploit strong sites, switch strategically, and preserve favorable responsiveness.
- This is the key reason the environment is an MDP and not a bandit.

Algorithm comparison and stability

Representative boxplot for H3 plus overall assessment



Tail episodic return distributions for Q-Learning and Double Q are broadly similar in this showcased setting.

Hypothesis assessment assessment

H1 Supported	Higher separability improves convergence and and final return.
H2 Supported	Increasing switch cost reduces switching frequency.
H3 Mixed	Double Q does not show a dramatic stability advantage in this representative case.
H4 Supported	Value Iteration is a useful upper-bound benchmark.
H5 Supported	Policies learn sequential state-aware behavior.

Takeaways and limitations

What the project demonstrates and what remains simplified

Key takeaways

- The project successfully turns stimulation-site selection into a genuine reinforcement learning problem.
- State-dependent responsiveness creates temporal coupling between current actions and future rewards.
- Model-free agents approach the Value Iteration benchmark in favorable settings but do not fully match it.
- Reward separability and switching cost jointly determine how easy the control problem is to learn.

Limitations and next steps

- The transition and EEG response models are stylized rather than fit to clinical data.
- State and observation spaces are intentionally discrete to keep the project interpretable.
- A natural next step is to learn or calibrate transition probabilities from real EEG or stimulation-response data.
- Another extension is richer partial observability, which would recover a clinically motivated POMDP.

Bottom line

Optimal behavior is a sequence policy: choose where to stimulate now while managing how today's action changes tomorrow's reward.